

Body-Band Suit: Controlled Physical Body-Shape Variation for Per-Garment Virtual Try-On

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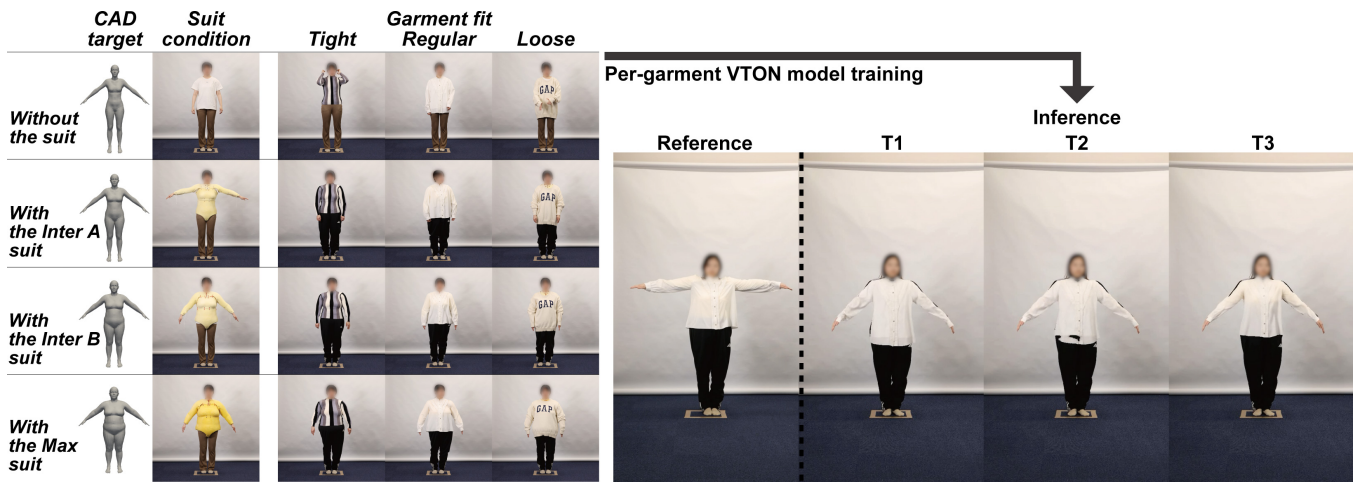


Figure 1: Overview of our virtual try-on (VTON) framework using Body-Band Suit. Left: Controlled physical data collection across four staged body-shape conditions (Base, Inter A, Inter B, and Max; rows) and three garment-fit conditions (Tight, Regular, and Loose; columns). The first two columns show the CAD target and suit condition. The arrow indicates per-garment VTON model training followed by inference. Right: Representative inference result for the *Inter B + Regular* condition, from left to right: real reference, T1 (trained with Base only), T2 (trained with Base + Max), and T3 (trained with Base + Inter A + Inter B + Max). In this example, T3 better preserves the garment silhouette and body coverage than T1 and T2.

Abstract

Image-based virtual try-on (VTON) systems aim to synthesize how a target garment would appear on a person, but evaluating and improving their robustness to body-shape variation remains difficult because body shape, garment behavior, pose, and capture conditions are usually entangled across different people and datasets. We present Body-Band Suit, a wearable apparatus that provides multiple staged body-shape conditions on the same participant using physically distinct suit configurations. Our approach complements synthetic body-shape manipulation and multi-subject data collection by varying apparent body shape on the same wearer. It

retains wearer-driven motion, physical garments, and real-world image capture without requiring a separate participant for each target condition. Using Body-Band Suit, we capture staged body-shape conditions ranging from a base body to intermediate and larger configurations, and construct training datasets with different levels of body-shape coverage. The current prototype follows a single staged enlargement trajectory derived from anthropometric references for Japanese women aged 20–29. We evaluate how these controlled body-shape conditions affect per-garment VTON results through quantitative metrics, qualitative comparisons, two diffusion-based baselines, and a perceptual user study. The results show that training-data configurations with broader staged body-shape coverage are associated with improved or maintained garment appearance and body coverage in several tested conditions, though the pattern varies with body shape and garment. These findings suggest that Body-Band Suit provides a practical way to collect training data and examine how per-garment VTON systems respond to changes in apparent body shape.



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CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Computer vision*; *Image processing*.

Keywords

virtual try-on, per-garment virtual try-on, body-shape variation, body-band suits, garment deformation

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1 Introduction

Image-based virtual try-on (VTON) systems aim to predict how garments appear on a person from visual inputs, but they still struggle to generalize across diverse body shapes [4, 6, 23]. This limitation becomes particularly evident for body-shape conditions that are underrepresented in the training data, where garment deformation and body coverage often become inconsistent [3, 15]. While recent methods have improved visual fidelity in common benchmark settings, robustness to body-shape variation remains limited [3, 14, 15].

A key challenge lies not only in model design, but also in data. Existing virtual try-on datasets often rely on naturally occurring body variation collected from human participants [6, 21] or on synthetic body manipulation and simulated shape changes [3, 14, 15]. As a result, it remains difficult to obtain systematically controlled body-shape conditions while preserving physical garment appearance, wearer-driven motion, and real-world image characteristics. Directly collecting garment images from participants with diverse body sizes is also costly and difficult to control, especially when matched garments, poses, and capture conditions are required. Consequently, the role of controlled body-shape coverage in training and evaluation remains insufficiently understood.

Recent work has lowered the barrier to collecting real-world per-garment virtual try-on data by using real human bodies instead of specialized robotic mannequins or more constrained capture setups [21]. However, in participant-based data collection, body-shape variation is still typically treated as an attribute of the recruited participants rather than as a controlled experimental variable. Moreover, while robotic or highly customized setups are attractive in principle, they remain limited in practice for reproducing diverse body shapes together with natural pose variation and wearer-driven garment deformation [1, 21]. Controlled body-shape variation on a human wearer can preserve freedom of movement while allowing data capture with matched garments and recording conditions, without requiring a separate participant for each staged condition.

To address this issue, we introduce *Body-Band Suit*, a wearable apparatus for controlled physical body-shape variation (Figure 1). The apparatus uses physically distinct suit configurations to provide an ordered set of discrete apparent body-shape conditions, ranging from a base body to intermediate and larger configurations. Rather

than replacing synthetic body-shape manipulation or the recruitment of participants with diverse body shapes, our approach complements these methods by enabling matched capture conditions on the same wearer. Using this setup, we construct three training-data configurations with progressively broader staged body-shape coverage. Because each additional captured condition also adds training frames, these configurations differ in both body-shape coverage and the amount of training data.

Specifically, T1 includes only the *Base* condition, T2 includes *Base* and *Max*, and T3 includes *Base*, *Inter A*, *Inter B*, and *Max*. We compare VTON performance across these training-data configurations. Across the quantitative metrics, no configuration performs best uniformly: broader staged coverage is associated with improved or maintained performance in several intermediate and larger body-shape settings, while T1 or T2 remains competitive in others. The qualitative comparisons and user study likewise reveal condition-dependent differences in garment silhouette, body coverage, and perceived similarity to real reference videos, including opposite preferences between garment-fit conditions in the T2–T3 comparison.

In summary, this paper makes the following contributions:

- (1) A wearable capture apparatus that provides a controlled set of discrete apparent body-shape conditions on the same wearer while retaining wearer-driven motion, physical garments, and matched capture conditions.
- (2) A staged dataset-design and evaluation framework comparing three training-data configurations under matched wearer, garment, and capture conditions.
- (3) A quantitative, qualitative, and perceptual analysis showing condition-dependent associations between broader staged body-shape coverage and per-garment VTON performance.

2 Related Work

2.1 Data-Driven Virtual Try-On

Data-driven VTON has made substantial progress in synthesizing garment appearance from a person image and a target garment image without explicit 3D garment reconstruction. Early image-based approaches such as CAGAN [10], VITON [6], and CP-VTON [18] established the basic paradigm of image-based garment transfer. Later methods improved preservation of body structure and clothing details, including VTNFP [26], ACGPN [24], and PF-AFN [4]. More recent methods further improved realism and controllability through stronger generative priors, including diffusion-based approaches such as TryOnDiffusion [27], LaDI-VTON [12], OOT-Diffusion [23], and ViViD [2]. Richer human representations such as DensePose [5] have also improved alignment between garments and body configuration.

Recent studies have begun to address garment fit explicitly. Fit-Controller introduces a fit-aware layout generator and a plug-in mechanism for controlling garment fit in existing VTON models [25]. FIT provides a large-scale synthetic dataset with body and garment measurements, together with a measurement-conditioned fit-aware baseline [20]. FitVTON learns from simulated garment-fitting data and conditions generation on body and garment size through structured descriptions [13]. These approaches address garment fit through model-side conditioning and fit-aware datasets

constructed from simulated or curated real-world image data. Our work addresses a complementary data-acquisition question: how to capture the same physical garment under staged apparent body-shape conditions on the same wearer for per-garment VTON training and evaluation.

2.2 Per-Garment Capture and Real-World Virtual Try-On Data Collection

A related approach improves fidelity by collecting garment-specific datasets and training per-garment VTON models. Chong et al. [1] proposed a per-garment capture and synthesis workflow based on an actuated mannequin and a measurement garment. Their method enables systematic capture of garment appearance under diverse body sizes and poses, and demonstrates that dedicated capture pipelines can better support rich interactions and fitting-related effects than generic image-based VTON. Subsequent work reduced the burden of capture through intermediate representations such as virtual measurement garments [22]. More recent approaches simplified dataset construction by using human wearers instead of robotic mannequins or other highly specialized capture systems [21]. These studies improve the practicality of per-garment data collection. Our study differs in emphasis by using wearable physical augmentation to introduce staged body-shape conditions while keeping the wearer, garment, and capture setup matched within each comparison.

2.3 Controlled Body-Shape Variation

Another relevant line of work studies how garment behavior depends on body shape, pose, and garment style. TailorNet [15], for example, models 3D clothing deformation as a function of pose, body shape, and garment style. Later work further addressed deformation of loose-fitting garments under articulated motion [14]. In the image-based setting, shape-controllable VTON has also been explored in more specialized scenarios such as underwear models [3]. Another relevant direction explores physical platforms for controllable body-shape change. For example, Takahashi and Koike [17] proposed a soft robotic body system that modulates body-fat-like characteristics through artificial muscles and fluidic control. While such systems are valuable as proof-of-concept platforms for physical body-shape manipulation, they are not designed for reproducing staged clothed body-shape conditions under practical real-world VTON capture conditions.

Together, these studies show that garment fit and body shape can be modeled, conditioned, simulated, or physically manipulated in complementary ways. Our goal is to provide a practical physical capture setup that introduces an ordered set of discrete apparent body-shape conditions along one predefined enlargement trajectory on the same wearer. Rather than replacing synthetic approaches or data collection involving participants with naturally diverse body shapes, Body-Band Suit serves as a complementary instrument for constructing and evaluating per-garment VTON training-data configurations under matched real-world capture conditions.

3 Body-Band Suits

3.1 Design Rationale

The goal of Body-Band Suit is to provide controlled and reproducible apparent body-shape variation on a human wearer. Instead of treating body shape as an uncontrolled attribute of recruited participants, we designed the suits as a wearable augmentation system that enables staged body-shape conditions while preserving the practical advantages of capture with a human wearer, i.e., natural pose freedom, wearer-driven motion, and physical garment deformation during actual wear.

This design addresses a key limitation of both subject-based and hardware-based alternatives. Recruiting participants across many body sizes makes it difficult to keep subject identity, garment, and capture conditions tightly matched. In contrast, robotic or rigid hardware-based systems can provide controllability, but may constrain wearer-driven motion or require specialized mechanisms to reproduce clothed silhouettes across diverse pose changes. Our aim was therefore to create a practical system that enables controlled body-shape variation while remaining compatible with the motion of a real human subject.

A further design requirement was that the resulting dressed appearance should remain plausible. The suits were therefore designed not merely to increase volume, but to produce apparent body-shape changes that yield plausible clothed silhouettes under the garments and poses examined in this study. The suits were intended as physical proxies for staged apparent body-shape variation in real-world VTON data collection.

3.2 Staged Body-Shape Design

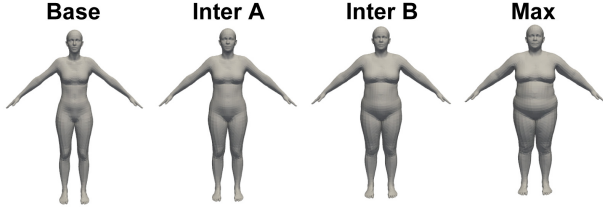
Using the Body-Band Suits, we defined four discrete apparent body-shape conditions sampled along one predefined enlargement trajectory: *Base*, *Inter A*, *Inter B*, and *Max*. The *Base* condition corresponds to the wearer without augmentation, while the other three conditions provide progressively larger apparent body shapes.

The staged conditions were derived using SMPL body-shape interpolation [11]. We first defined a reference-body endpoint and a maximum-body endpoint using two different HQL resources. The reference endpoint was based on the licensed individual homologous body-model data for Japanese women aged 20–29 from the HQL Japanese Body Dimension Database 2004–2006 [9], whereas the target measurements for the larger endpoint were based on the HQL Japanese Body Measurement Databook 2004–2006 [8]. We then uniformly interpolated the corresponding SMPL shape parameters between these two endpoints and selected two intermediate parameter points, denoted *Inter A* and *Inter B*. The samples are evenly spaced in SMPL parameter space; this does not imply equal changes in physical dimensions, perceived body size, or biomechanical properties.

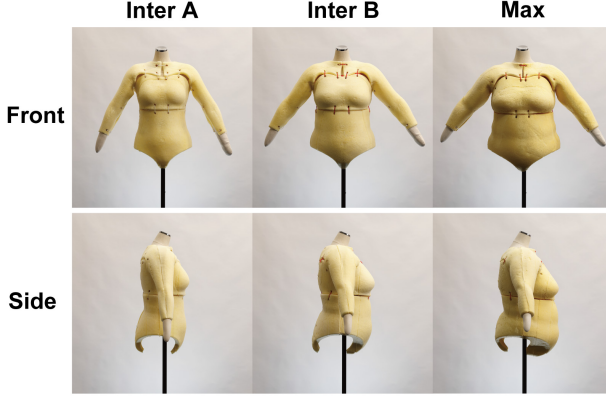
Figure 2 shows the resulting staged body-shape models together with the fabricated suits and an example of the mold-based fabrication process.

This staged design is central to our study. If only *Base* and *Max* are available, the training data represent body-shape change only as a binary contrast. Introducing intermediate conditions allows us to compare training-data configurations with different levels of staged body-shape coverage. Because each additional captured

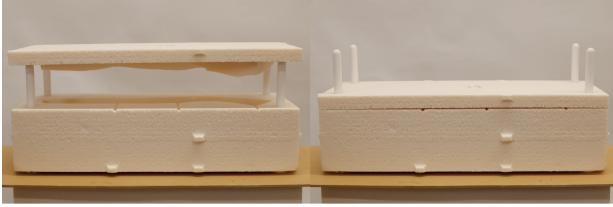
(a) SMPL shape design



(b) Fabricated suits



(c) Mold for arm-front component



(d) Cast arm-front component

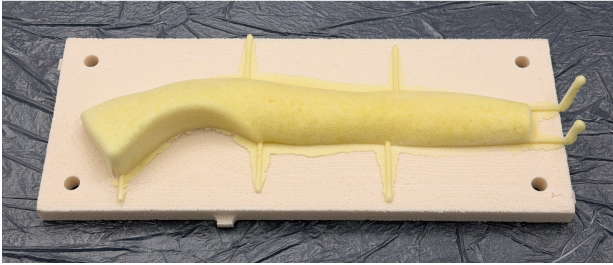


Figure 2: Body-Band Suits and their staged body-shape design. (a) SMPL-based body-shape models used to define the four staged conditions: Base, Inter A, Inter B, and Max. (b) Fabricated Body-Band Suits for Inter A, Inter B, and Max, shown from the front and side. (c) Example of a CNC-machined polystyrene foam mold used to fabricate an arm-front component. (d) Cast polyurethane arm-front component.

condition also adds training frames, however, this comparison does not isolate the causal effect of intermediate body-shape coverage

Table 1: Representative SMPL/CAD-derived target dimensions (cm) for the four staged body-shape conditions.

Measurement	Base	Inter A	Inter B	Max
Chest	83.8	96.5	109.4	122.5
Hip	93.3	101.7	109.4	116.8
Waist	69.9	82.7	96.0	109.7
Shoulder breadth	31.0	32.7	34.4	36.0

from that of additional training data. In this sense, the term *body-band* refers not only to the physical construction of the suits, but also to the use of intermediate conditions to represent body-shape change as an ordered staged progression rather than an isolated endpoint comparison.

Table 1 summarizes representative target dimensions used to design the four staged conditions. These values are derived from the SMPL/CAD models, not from measurements of the fabricated suits or wearers in the suits. The progression from *Base* to *Max* can be seen in the chest, hip, waist, and shoulder breadth dimensions, while *Inter A* and *Inter B* provide intermediate design targets between the two endpoints. Hip dimensions are included because the augmented geometry extends through the pelvis toward the upper-thigh region.

3.3 Fabrication and Structure

Each Body-Band Suit was fabricated as a set of custom polyurethane foam parts cast from CNC-machined molds. We first milled mold blocks from polystyrene foam using a large-format CNC machine. The mold cavities were designed in CAD and corresponded to the negative volume of the target body-augmentation geometry. Liquid expanding polyurethane foam was then poured into the mold cavities to produce the suit components.

A key technical challenge is that polyurethane foam typically adheres strongly to polystyrene foam molds, making stable demolding difficult and degrading both dimensional accuracy and surface quality. To address this, we developed a low-cost mold-treatment workflow using a two-layer coating: a silicone-based coating applied to the machined polystyrene surface, followed by a mold-release layer. This combination enabled repeated demolding with more consistent surface quality while avoiding the cost and fabrication time required for full silicone molds or metal tooling. Figure 2(c) and (d) show an example of the mold and the resulting cast component.

Each suit consists of eight cast parts in total: four arm parts (two for each arm), two chest parts, and two abdominal parts. For each body region, the front and back volumes were cast separately and then bonded together, forming a volumetric slice of the corresponding body segment. After bonding, the parts were organized into four wearable units: left arm, right arm, chest, and abdomen. These units were then connected by laces to form the complete suit. Figure 2(b) shows the fabricated suits for *Inter A*, *Inter B*, and *Max*.

The inner cavity of each suit was derived from the reference-body geometry. The design therefore requires a wearer whose body proportions are reasonably close to the reference and who fits inside all three augmented suits. The same wearer can then be used across the *Base*, *Inter A*, *Inter B*, and *Max* conditions, avoiding the need

to recruit a different participant for each target condition while keeping wearer identity fixed. The compliance of the polyurethane foam and the adjustable lace closures accommodate moderate differences between the wearer and the reference geometry, but are not intended to fit arbitrary wearers or make different wearers geometrically identical.

The augmented volume covers the upper body as a whole, extending from the base of the neck to the wrists, and from the torso down to the upper-thigh region around the pelvis. Thus, the suit modifies not only the front torso but also the back, lateral torso, pelvis region, and both arms, allowing body-shape change to appear as a whole upper-body silhouette rather than as isolated local padding.

The suit opens along the underside of the arms and along the back, approximately following the spinal line. The suit is placed on the wearer and secured by tightening the laces. This design enables relatively simple donning and doffing while maintaining stable fixation during motion. The lace-based closure is used to secure the wearable units and reduce displacement during capture.

3.4 Practical Properties for Data Collection

The Body-Band Suits were designed not only to alter apparent body shape, but also to remain compatible with upper-body motion during data collection. The parts were intentionally designed as continuous volumes across major joints, especially around the shoulder, rather than segmented at every joint. This reduces unnatural local bulging during arm motion and helps preserve a plausible clothed silhouette during pose changes.

The compliance of the polyurethane foam permits limited deformation as the wearer moves, allowing the augmented volume to follow wearer-driven motion. The material was not intended or validated to reproduce the mechanical response of human soft tissue. Instead, it provides a compliant physical support beneath the garment while remaining compatible with the pose variation used during capture.

Another practical advantage is reproducibility. Conventional mold-based fabrication of soft volumetric parts often requires large amounts of silicone or dedicated tooling, which increases both cost and fabrication time. By contrast, our mold workflow uses CNC-machined polystyrene foam together with a coating and release process, enabling repeated fabrication at lower cost while maintaining sufficient dimensional stability for the staged experimental conditions. This makes the suits not only wearable, but also practically reproducible as an experimental platform.

The Body-Band Suits thus provide an experimental instrument for controlling apparent body shape along the defined enlargement trajectory during real-world per-garment VTON data collection.

4 Dataset and Training Setup

4.1 Overview

Our goal is to examine how staged body-shape coverage in training data is associated with per-garment VTON performance. To this end, we constructed a garment-specific dataset under matched real-world capture conditions using Body-Band Suit. We kept the garment synthesis architecture and optimization settings fixed and varied the body-shape conditions included in the training data.

We organized the captured data into three training-data configurations with progressively broader staged body-shape coverage. Because each additional body-shape condition also adds captured frames, the configurations differ in both body-shape coverage and dataset size. Therefore, their comparison does not isolate the causal effect of intermediate body-shape coverage from that of additional training data.

4.2 Base Synthesis Framework

We build on the real-human per-garment VTON framework of Wu et al. [21]. The framework represents a person using a virtual measurement garment (VMG) image and a simplified DensePose map. Given an input frame, BEV [16] estimates an SMPL body mesh, from which the VMG image is rendered. DensePose [5] provides image-space alignment information. The two three-channel images are concatenated to form a six-channel hybrid person representation.

The garment synthesis (GS) network takes this hybrid representation as input and predicts both the target garment image and its mask. The GS network and discriminator are based on pix2pixHD [19]. During inference, the predicted garment image and mask are transformed from the upper-body region of interest (ROI) back to the original image coordinates and composited with the input frame. Figure 3 summarizes the training and inference pipeline used in this study.

4.3 Body-Shape and Garment Conditions

We defined four ordered, discrete body-shape conditions: *Base*, *Inter A*, *Inter B*, and *Max*. The *Base* condition corresponds to the wearer without a suit, while the other conditions use the three Body-Band Suits described in Section 3. These conditions provide staged apparent body-shape variation within the same physical capture setup.

We used three upper-body garments representing different relative fit conditions: *Tight*, *Regular*, and *Loose*. These labels describe the observed fit of the garments on the training wearer rather than standardized commercial size categories. The *Tight* garment closely follows the torso, the *Regular* garment provides moderate ease, and the *Loose* garment provides greater ease and visible drape. This allows us to compare the training-data configurations across garments with different fits.

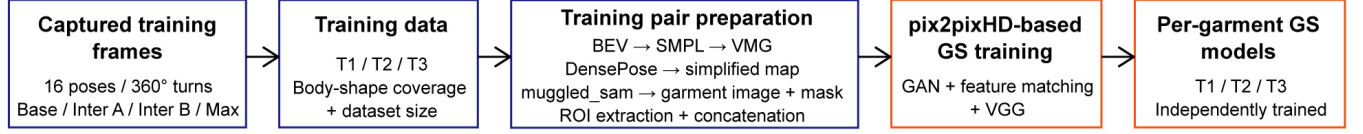
4.4 Data Collection

One wearer provided the training data for all four body-shape conditions and all three garments. The training wearer was 152 cm tall, with chest, waist, and hip circumferences of 77, 67, and 85 cm, respectively; body weight was not reported.

For each garment and body-shape condition, the training wearer performed a predefined motion sequence while being recorded at 24 fps. Each recording lasted approximately two minutes and contained approximately 2,880 frames. The sequence consisted of 16 predefined pose patterns, extending the 14-pose procedure of Wu et al. [21]. For each pose, the wearer performed one full 360° turn so that garment appearance could be captured from multiple viewing directions. Figure 4 shows the 16 poses using the *Base* condition for clarity.

No temporal frame sampling or manual frame removal was performed, and the same preprocessing procedure was used for T1, T2,

(a) Training



(b) Inference

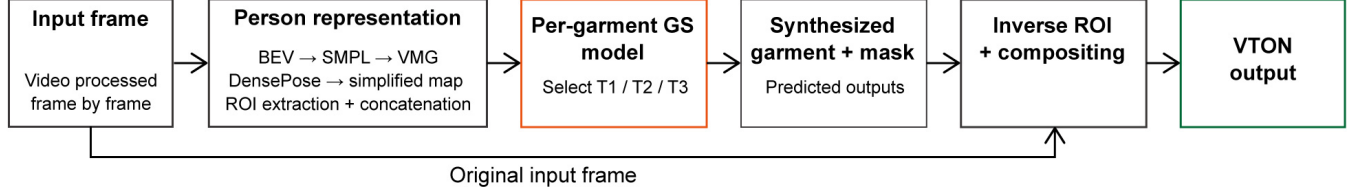


Figure 3: Overview of the training and inference pipeline. (a) Captured frames are organized into the T1, T2, and T3 training-data configurations. For each frame, BEV-based SMPL estimation and DensePose produce the VMG image and simplified DensePose map, while garment segmentation produces the target garment image and mask. These paired data are used to train a pix2pixHD-based per-garment GS model for each configuration. (b) During inference, the hybrid person representation extracted from each input frame is processed by the selected GS model. The predicted garment image and mask are transformed back to the original image coordinates and composited with the input frame to produce the VTON output.

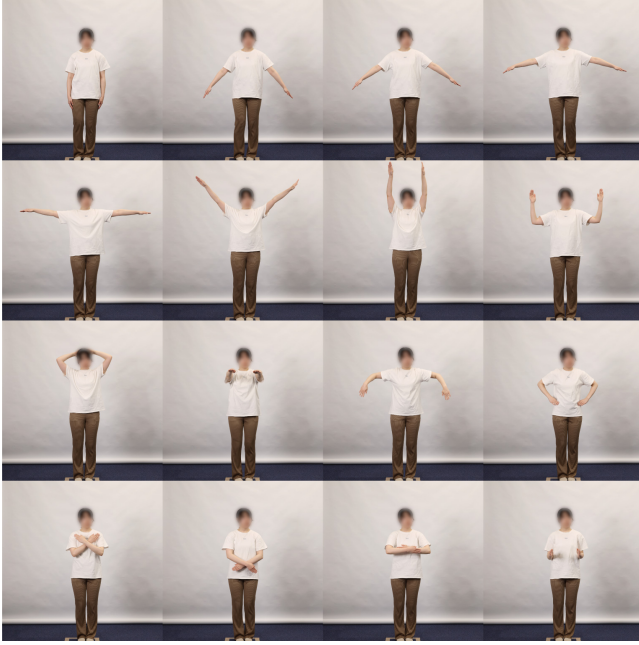


Figure 4: The 16 predefined pose patterns used for data collection. For each pose, the training wearer performed one full 360° turn while wearing the target garment under a fixed body-shape condition. The poses are shown using the *Base* condition for clarity.

and T3. We did not use a separate validation split. At 24 fps, these approximate recording durations correspond to about 2,880, 5,760, and 11,520 frames per garment for T1, T2, and T3, respectively. These counts make explicit that adding body-shape conditions also increases the amount of training data.

4.5 Per-Garment Dataset Construction

The recorded videos were processed frame by frame following the pipeline of Wu et al. [21], with muggled_sam replacing Graphonomy for garment segmentation in the implementation used for this study. BEV was used to estimate an SMPL mesh and generate the VMG image, while DensePose was used to produce the simplified DensePose map. Garment masks were generated using muggled_sam [7], and each mask was applied to the corresponding recorded frame to obtain the target garment image.

The upper-body ROI was extracted and resized to 512×512 pixels. The VMG image and simplified DensePose map were concatenated as the network input, while the target garment image and mask served as the training targets. Following Wu et al., random affine transformations were applied after ROI extraction for data augmentation.

As in previous per-garment VTON workflows, a separate GS model was prepared for each target garment. We did not add an explicit body-shape feature to the person representation; body-shape information enters the experiment through the physically captured conditions and through the SMPL- and DensePose-based representations extracted from each frame. Because ground-truth SMPL parameters were unavailable, the accuracy of the fitted SMPL meshes was not quantitatively evaluated in this study. The potential effect of this unmeasured fitting error is discussed in the limitations.

4.6 Training-Data Configurations

We constructed the following three training-data configurations:

- **T1:** *Base* only.
- **T2:** *Base* and *Max*.
- **T3:** *Base*, *Inter A*, *Inter B*, and *Max*.

T1 represents the base-only setting. T2 adds the largest staged body-shape condition, and T3 includes both intermediate conditions in addition to the two endpoints. The T2–T3 comparison

therefore shows what happens when intermediate conditions and their associated training frames are added; it does not separate the effects of intermediate coverage and dataset size.

For each of the three garments, GS models were trained for T1, T2, and T3, resulting in nine models in total. All GS models were initialized and trained independently from scratch, without using pretrained pix2pixHD or Wu et al. checkpoints. No checkpoints were transferred among T1, T2, and T3. This avoids carrying over learned garment-specific weights between training-data configurations.

4.7 Implementation Details

Following Wu et al. [21], the GS network and discriminator were derived from pix2pixHD. The training objective combined the adversarial loss, feature-matching loss, and VGG perceptual loss. The coefficients for the feature-matching and VGG perceptual losses were both set to 1.

All models were trained for 100 epochs with a batch size of 8 and a learning rate of 2×10^{-4} . We used the Adam optimizer with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. Training was performed on an NVIDIA RTX A6000 GPU. The ROI resolution was 512×512 pixels for all configurations. The architecture and optimization settings were kept unchanged across T1, T2, and T3.

4.8 Evaluation Data and Protocol

The final quantitative evaluation used one wearer who did not contribute to the training data, hereafter referred to as the evaluation wearer. The evaluation wearer was 158.5 cm tall and weighed 45.5 kg; chest, waist, and hip measurements were not reported. All quantitative results reported in this paper use this same evaluation wearer.

For each evaluated body-shape and garment condition, the synthesized video was compared with a real reference video showing the evaluation wearer physically wearing the corresponding garment under the corresponding body-shape condition. We evaluated all combinations of the four body-shape conditions and three garment conditions when comparing T1, T2, and T3.

We use quantitative metrics, qualitative comparisons, and a perceptual user study as complementary forms of evaluation. The quantitative evaluation uses SSIM, LPIPS, and VFID to measure frame-level structural similarity, perceptual similarity, and video-level distributional similarity. These metrics do not directly measure garment-fit accuracy, so garment silhouette, body coverage, and visible artifacts are also examined qualitatively.

For contextual comparison, we additionally evaluate OOTDiffusion [23] and ViViD [2] under four selected intermediate and larger-body conditions. Both methods were applied zero-shot, without garment-specific training, using the same input and reference videos as T1, T2, and T3. In contrast, our T3 models were trained separately for the corresponding garments. Consequently, this comparison is not treated as a controlled ranking of model architectures, but as a contextual reference showing how general-purpose VTON methods perform on the same evaluation samples.

5 Quantitative Evaluation

5.1 Evaluation Metrics

We conducted a direct quantitative evaluation using paired real reference videos, i.e., videos of the evaluation wearer physically wearing the corresponding garment under each staged body-shape condition. Each synthesized VTON video was compared with its corresponding real reference video.

We used three complementary metrics: SSIM, LPIPS, and VFID. SSIM measures frame-wise structural similarity to the reference. LPIPS measures frame-wise perceptual distance in deep feature space and is less sensitive to small pixel-level misalignment than pixel-based metrics. VFID measures distributional similarity between synthesized and reference videos using video-level features, capturing both visual appearance and temporal behavior. Higher SSIM and lower LPIPS and VFID indicate closer agreement with the references.

These metrics do not directly measure garment-fit accuracy, garment silhouette, or body coverage. We therefore interpret the quantitative results together with the qualitative comparisons and user-study results.

5.2 Comparison of Training-Data Configurations

Table 2 compares T1 (Base only), T2 (Base + Max), and T3 (Base + Inter A + Inter B + Max). Across the 12 body-shape and garment combinations and the three metrics, T3 gives the numerically best or tied-best value in 24 of the 36 entries.

The strongest pattern appears under the intermediate body-shape conditions. For *Inter A*, T3 is best or tied-best in eight of the nine entries, including every LPIPS and VFID value. For *Inter B*, T3 is best or tied-best in seven of the nine entries, including all three metrics for the Regular garment. Under *Inter B + Loose*, T2 and T3 tie in SSIM, T3 has a slightly lower LPIPS value, and T2 has a slightly lower VFID value. Thus, the difference between T2 and T3 in this condition is small and metric-dependent.

The endpoint conditions show a more mixed pattern. Under *Base*, T3 is best or tied-best in six of the nine entries, while T1 and T2 retain the best values for some Tight and Regular settings. Under *Max*, T2 gives the best value in five of the nine entries, whereas T3 gives the best value for all three metrics for the Loose garment. Therefore, no configuration performs best uniformly across all conditions.

The numerical results favor T3 most often, particularly under the two intermediate body-shape conditions. This result supports the practical value of the complete staged training-data configuration evaluated in this study. However, because T1, T2, and T3 contain different numbers of training frames, the present comparison does not determine whether this advantage comes from broader staged body-shape coverage, additional training data, or both.

5.3 Comparison with External Baselines

We additionally compared T3 with two recent diffusion-based baselines, OOTDiffusion [23] and ViViD [2], under four selected conditions: *Inter B + Tight*, *Inter B + Regular*, *Max + Tight*, and *Max + Regular*. These conditions cover the *Inter B* and *Max* body-shape

Table 2: Quantitative comparison of the three training-data configurations: T1 (Base only), T2 (Base + Max), and T3 (Base + Inter A + Inter B + Max). Higher SSIM is better, whereas lower LPIPS and VFID are better. Best values in each row are shown in bold.

Input	Garment	T1			T2			T3		
		SSIM↑	LPIPS↓	VFID↓	SSIM↑	LPIPS↓	VFID↓	SSIM↑	LPIPS↓	VFID↓
Base	Tight	0.8440	0.2092	0.7018	0.8438	0.2094	0.7168	0.8493	0.2092	0.7108
	Regular	0.8608	0.2043	0.7024	0.8608	0.2040	0.6970	0.8610	0.2046	0.7157
	Loose	0.8594	0.2067	0.7644	0.8565	0.2094	0.7663	0.8600	0.2053	0.7535
Inter A	Tight	0.8456	0.2119	0.6805	0.8450	0.2125	0.7054	0.8453	0.2118	0.6702
	Regular	0.8637	0.2055	0.6243	0.8640	0.2051	0.6134	0.8649	0.2038	0.6004
	Loose	0.8556	0.2064	0.6678	0.8548	0.2071	0.6576	0.8568	0.2043	0.6547
Inter B	Tight	0.8423	0.2194	0.6355	0.8422	0.2196	0.6278	0.8422	0.2193	0.6208
	Regular	0.8587	0.2142	0.6446	0.8601	0.2124	0.6162	0.8606	0.2115	0.6060
	Loose	0.8054	0.2321	0.9194	0.8079	0.2283	0.8845	0.8079	0.2281	0.8881
Max	Tight	0.8370	0.2036	0.6477	0.8332	0.2024	0.6382	0.8330	0.2027	0.6397
	Regular	0.8460	0.2025	0.6785	0.8504	0.1974	0.6080	0.8484	0.2007	0.6196
	Loose	0.8410	0.2084	0.6718	0.8441	0.2033	0.5851	0.8449	0.2022	0.5850

Table 3: Contextual quantitative comparison with two zero-shot external baselines under four selected conditions; T3 is trained per garment. Higher SSIM is better, whereas lower LPIPS and VFID are better. Best values in each row are shown in bold.

Input	Garment	OOTDiffusion			ViViD			Ours (T3)		
		SSIM↑	LPIPS↓	VFID↓	SSIM↑	LPIPS↓	VFID↓	SSIM↑	LPIPS↓	VFID↓
Inter B	Tight	0.8305	0.2183	0.8873	0.8106	0.2687	1.0702	0.8422	0.2193	0.6208
Inter B	Regular	0.8472	0.2116	0.9663	0.8341	0.2515	1.1354	0.8606	0.2115	0.6060
Max	Tight	0.8312	0.2054	0.9225	0.7907	0.2212	1.0151	0.8330	0.2027	0.6397
Max	Regular	0.8482	0.2012	0.9637	0.8241	0.2679	1.2681	0.8484	0.2007	0.6196

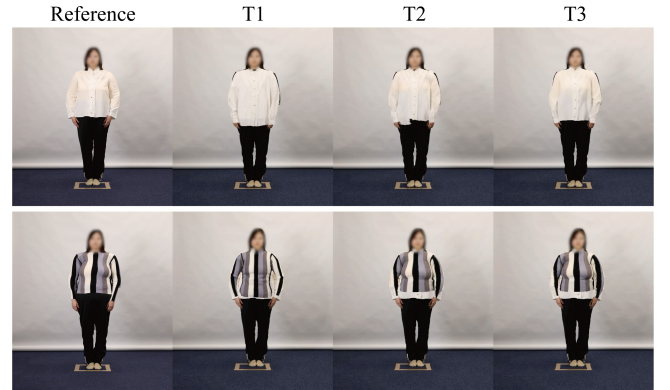
stages for the Tight and Regular garments. OOTDiffusion and ViViD were applied zero-shot, using the same source videos and real reference videos as T3, whereas each T3 model was trained for the corresponding garment.

Table 3 summarizes the results. T3 gives the numerically best value in 11 of the 12 metric entries. It achieves the highest SSIM and lowest VFID in all four conditions and the lowest LPIPS in three conditions. The only exception is *Inter B + Tight*, for which OOTDiffusion gives a slightly lower LPIPS value than T3 (0.2183 versus 0.2193). The consistently lower VFID values are the clearest numerical difference in this comparison, indicating closer agreement with the real reference videos at the video-feature distribution level.

Because the training setups differ, these results provide a contextual comparison rather than a controlled ranking of model architectures. The observed differences may reflect garment-specific training and its associated captured data, in addition to differences between the synthesis methods. The comparison therefore describes performance on the selected samples without establishing the superiority of the synthesis architecture.

6 Qualitative Evaluation

Figure 5 shows representative qualitative comparisons among T1, T2, and T3 under the *Max* body-shape condition for two garments. The top row shows *Max + Regular*, and the bottom row shows *Max*

**Figure 5: Representative qualitative comparisons under the *Max* body-shape condition for the Regular (top) and Tight (bottom) garments. From left to right: real reference, T1 (Base only), T2 (Base + Max), and T3 (Base + Inter A + Inter B + Max). The examples show differences in overall garment silhouette, hemline continuity, and garment extent over the lower torso.**

+ *Tight*. From left to right, each row presents the real reference, T1, T2, and T3.

The most visible differences appear in the overall garment silhouette, hemline continuity, and garment extent over the lower

torso. In both rows, T3 more closely matches the reference than T1 in overall silhouette and garment extent. Under *Max + Regular*, T2 shows a visible dark discontinuity near the center of the hem, whereas T3 maintains a continuous hemline closer to the reference. Under *Max + Tight*, the differences between T2 and T3 are smaller: both preserve the body-following striped contour, whereas T1 shows larger deviations around the torso and sleeves. Thus, the qualitative difference between T2 and T3 is clear in the top row but limited in the bottom row.

These examples complement the quantitative results by showing local differences in silhouette, hemline continuity, and garment extent that are not directly measured by SSIM, LPIPS, or VFID. They also reinforce that the differences among the three training-data configurations are condition-dependent rather than uniform. Because the figure presents selected frames, these observations are interpreted together with the full-video metrics and user-study results.

7 User Study

7.1 Study Design

To complement the quantitative and qualitative evaluations, we conducted two exploratory perceptual studies comparing synthesized VTON videos with real reference videos. We separately assessed three judgments: similarity to the reference, perceived naturalness, and agreement with the body coverage shown in the reference. These criteria were treated separately because they represent different aspects of the perceived result.

Study 1 compared T1 and T3 under four conditions: *Inter B + Tight*, *Inter B + Regular*, *Max + Tight*, and *Max + Regular*. Study 2 compared T2 and T3 under two conditions: *Inter B + Tight* and *Inter B + Regular*. Study 1 therefore compares the base-only and complete staged training-data configurations, whereas Study 2 compares the base-plus-maximum configuration with the complete staged configuration. Because T2 and T3 also contain different numbers of training frames, Study 2 does not isolate the causal effect of adding intermediate body-shape conditions.

We recruited 25 adult participants from a university community. Expertise in virtual try-on, computer vision, or garment design was not required because the study was intended to capture general perceptual judgments.

The study was implemented as an online questionnaire. In each comparison, participants were shown one real reference video and two synthesized result videos. The A/B assignment of the two synthesized videos was counterbalanced across two questionnaire versions. Responses were subsequently mapped back to the corresponding model labels, and *About the same* responses were retained as neutral responses.

For each comparison, participants answered three questions: which result was closer to the reference, which looked more natural as a video of a person wearing the garment, and which better matched the body coverage shown in the reference. Each question used a three-way response: A, B, or *About the same*.

Because the study was exploratory, we report descriptive response counts and preference rates without making claims of statistical significance. Neutral responses were retained in the main counts, and preference rates excluding neutral responses were used

only to summarize directional tendencies. Within each condition, the pooled counts combine three answers from each participant. They summarize preferences without treating the three criteria as interchangeable or the responses as independent observations.

Ethics, consent, and compensation. The wearer-capture protocol was reviewed and approved by the institutional ethics committee at The University of Tokyo. The recorded wearers provided written informed consent and were compensated for their participation. The subsequent perceptual study was an uncompensated online questionnaire in which adult participants viewed the reference and synthesized videos and provided three-choice responses; it did not undergo separate ethics review.

7.2 Results

Figure 6 visualizes the overall response distributions, and Table 4 reports the corresponding counts. In Study 1, T3 was preferred over T1 for all three questions after neutral responses were excluded. The T3 preference rates were 57.6% for similarity to the reference, 62.5% for naturalness, and 60.0% for body coverage. These descriptive results show an overall perceptual preference for T3 over T1, with the highest preference rate observed for naturalness.

The condition-wise counts in Table 5 show that this preference was not uniform. The clearest preference for T3 appeared under *Max + Tight*, with 42 responses for T3, 18 for T1, and 15 neutral responses when the three questions were pooled. T3 was also preferred under *Max + Regular* by 30 responses to 17, with 28 neutral responses. The *Inter B* conditions were more mixed: *Inter B + Tight* favored T3 by 28 responses to 20, whereas *Inter B + Regular* favored T1 by 25 responses to 20. These counts indicate that the overall Study 1 tendency was driven mainly by the two *Max* conditions.

Study 2 shows a stronger reversal between garment conditions. Although the overall preference rates excluding neutral responses favored T3 for all three questions, the pooled condition-wise counts followed opposite directions. Under *Inter B + Tight*, most responses were neutral (40 of 75); among the remaining responses, T2 was selected 30 times and T3 five times. Under *Inter B + Regular*, T3 was selected 48 times, compared with five selections for T2 and 22 neutral responses. Thus, the overall Study 2 result is driven by *Inter B + Regular* and does not represent a uniform preference for T3 over T2.

This reversal provides additional context for the quantitative results. Under *Inter B + Regular*, T3 gives numerically better results than T2 for all three metrics, consistent with the strong perceptual preference for T3. Under *Inter B + Tight*, T2 and T3 have the same SSIM, and T3 has only slightly lower LPIPS and VFID values, while the non-neutral user-study responses favor T2. This difference shows that the quantitative metrics and the three perceptual questions do not necessarily produce the same ordering. The present study does not identify which specific local visual cues caused these judgments.

Overall, non-neutral responses favored T3 over T1, particularly under the two *Max* conditions, while the T2–T3 preference varied with garment condition. These descriptive findings concern the complete training-data configurations; they do not separate the contributions of intermediate body-shape coverage and additional training data.

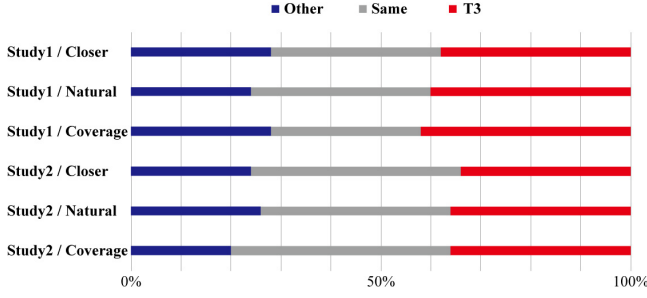


Figure 6: Overall response distributions for Study 1 (T1 vs. T3) and Study 2 (T2 vs. T3). Other denotes T1 in Study 1 and T2 in Study 2. Each bar shows the percentage of responses selecting the compared model, *About the same*, or T3.

8 Conclusion

We introduced Body-Band Suit, a wearable apparatus for creating controlled and reproducible apparent body-shape conditions on the same wearer. Using physically distinct suit configurations, the setup covers a predefined progression from a base body to intermediate and larger shapes while preserving wearer-driven motion, physical garments, and matched capture conditions. This provides a practical way to treat apparent body shape as an explicit experimental variable in per-garment VTON data collection and evaluation.

Using this setup, we compared three training-data configurations: T1 (Base only), T2 (Base + Max), and T3 (Base + Inter A + Inter B + Max). T3 achieved the best or tied-best values most often, particularly under the intermediate conditions, but no configuration was uniformly best. Qualitative comparisons likewise showed condition-dependent differences in garment silhouette, hemline continuity, and coverage. Because the configurations differ in both staged body-shape coverage and training frame count, the observed differences cannot be attributed to body-shape coverage alone.

In the exploratory user study, non-neutral responses favored T3 over T1 overall, particularly under the Max conditions, whereas the T2–T3 preference reversed between the Tight and Regular garments under Inter B. Perceptual preferences thus depended on the garment condition.

In practice, training-data configurations should be assessed for the target garment and body-shape conditions. Numerical scores and perceptual judgments provide complementary evidence for this assessment; an overall ranking can obscure differences that matter for a particular case. Body-Band Suit provides a way to collect real reference videos for these comparisons across planned body-shape conditions while keeping wearer identity fixed.

The study is limited to one predefined SMPL/CAD-derived enlargement trajectory based on anthropometric references for Japanese women aged 20–29, one training wearer, one evaluation wearer, and three garments. The polyurethane suits are controlled physical proxies rather than biomechanical replicas of human soft tissue, and validation on naturally larger-bodied participants remains future work. SMPL fitting accuracy was not quantified; fitting errors may attenuate differences among the suit conditions. The external comparison is contextual because garment-specific T3 models were compared with zero-shot general-purpose methods. Body-Band Suit

Table 4: Overall user-study results. Study 1 compares T1 with T3, and Study 2 compares T2 with T3. Other denotes T1 in Study 1 and T2 in Study 2. Counts show responses selecting the compared model, T3, or *About the same*. The last column reports the T3 preference rate after excluding neutral responses.

Study	Question	Other	T3	Same	T3 pref.
Study 1	Closer	28	38	34	57.6%
	Natural	24	40	36	62.5%
	Coverage	28	42	30	60.0%
Study 2	Closer	12	17	21	58.6%
	Natural	13	18	19	58.1%
	Coverage	10	18	22	64.3%

Table 5: Condition-wise response counts pooled across the three questions. Other denotes T1 in Study 1 and T2 in Study 2. The pooled counts provide a directional summary and do not imply that the three perceptual criteria are equivalent.

Study	Condition	Other	T3	Same
Study 1	Inter B + Tight	20	28	27
	Inter B + Regular	25	20	30
	Max + Tight	18	42	15
	Max + Regular	17	30	28
Study 2	Inter B + Tight	30	5	40
	Inter B + Regular	5	48	22

should complement the recruitment of participants with naturally diverse body shapes.

Future work should use frame-count-matched ablations and extend the evaluation to a broader range of participants, body shapes, and garments. An actively reconfigurable suit could allow continuous adjustment of apparent body shape within a defined range without exchanging suits, while retaining wearer-driven motion. Such control could support denser sampling along body-shape trajectories and richer training data on garment appearance across shapes and poses. The suit could also be adapted to a general-purpose humanoid robot, with the suit providing body-shape variation and the robot providing motion. Both applications would require maintaining plausible garment contact and deformation across body-shape and pose changes.

Body-Band Suit allows apparent body shape to be varied systematically while keeping the wearer and capture setup fixed. It provides a practical basis for studying how real garments respond to staged body-shape changes and how faithfully VTON systems reproduce their appearance during motion.

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